

# MID TERM EXAMINATION

## Subject: computer networks

Paper: Mid Term Examination - Set A | Duration: 3 Hours | Max Marks: 28 | Date: 13-03-2026

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Total Questions: 10 | Answer all questions.

Name: \_\_\_\_\_

Roll No: \_\_\_\_\_

Student Sign: \_\_\_\_\_

Teacher Sign: \_\_\_\_\_

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### SECTION A: MCQS

1. Which of the following best describes an Abstract Data Type (ADT)? [2 Marks]
  - a) A data structure that stores primitive data types only.
  - b) A concrete implementation of a data structure.
  - c) A mathematical model that specifies the data stored and the operations that can be performed on that data, without specifying its implementation.
  - d) A data type that can only be accessed sequentially.
  
2. What is the primary disadvantage of using an array-based implementation for a stack or queue compared to a linked list implementation? [2 Marks]
  - a) Slower access time for elements.
  - b) Difficulty in implementing basic operations like push/pop or enqueue/dequeue.
  - c) Fixed size limitation, leading to potential overflow or underflow issues if not dynamically resized.
  - d) Higher memory consumption due to pointers.
  
3. In the context of binary trees, what is the 'height' of a tree? [2 Marks]
  - a) The total number of nodes in the tree.
  - b) The length of the longest path from the root node to any leaf node.
  - c) The maximum number of children any node has.
  - d) The number of levels in the tree excluding the root level.
  
4. Which of the following is NOT a linear data structure? [2 Marks]
  - a) Linked List
  - b) Stack
  - c) Queue
  - d) Binary Tree
  
5. Consider an algorithm with a time complexity of  $O(n^2)$ . If the input size 'n' doubles, how does the execution time approximately change? [3 Marks]
  - a) It doubles.
  - b) It quadruples.
  - c) It remains the same.
  - d) It increases logarithmically.
  
6. In a singly linked list, which of the following operations has a time complexity of  $O(1)$  in the worst case, assuming a pointer to the head of the list is available?

pointer to the head node is available?

[3 Marks]

- a) Accessing the k-th element.
- b) Searching for a specific element.
- c) Deleting the last element.
- d) Inserting a new node at the beginning of the list.

7. Which of the following data structures is most suitable for implementing a "undo" functionality in a text editor?[3 Marks]

- a) Queue
- b) Array
- c) Stack
- d) Hash Table

8. What is the worst-case time complexity for inserting an element into a Binary Search Tree (BST) of height 'h'?[3 Marks]

- a)  $O(1)$
- b)  $O(\log h)$
- c)  $O(h)$
- d)  $O(h \log h)$

9. A circular queue with a maximum capacity of N elements is implemented using an array. If front points to the first element and rear points to the last element, which condition indicates that the queue is full?

[4 Marks]

- a) `front == rear`
- b) `front == (rear + 1) % N`
- c) `rear == front - 1`
- d) `front == 0 && rear == N - 1`

10. Given the following inorder and preorder traversals of a binary tree, what would be its postorder traversal?

Inorder: D B E A F C

Preorder: A B D E C F

[4 Marks]

- a) D E B F C A
- b) D B E A F C
- c) A B D E C F
- d) A D B E F C

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